

Stanford University

Department of Earth System Science
450 Jane Stanford Way, Bldg. 320 Braun Hall
Stanford, CA 94305

marcusnobrega.engcivil@gmail.com | +1 520 390-2813

[Website](#) | [Google Scholar](#) | [GitHub](#) | [ORCID](#)

Marcus N.
Gomes Jr.



Academic Background

- Feb. **Postdoctoral Appointment**, *Stanford University*, Concentration area: Earth
2026–present System Science, Floods and its relationship with medical records in India and
California
Advisor: Steve Gorelick, Ph.D.
- Jan. **Postdoctoral Appointment**, *University of Arizona*, Concentration area: Hy-
2024–Jan. drology and Atmospheric Sciences, NSF Project: Hydrologic closure relationships
2026 at different levels of hillslope model complexity
Advisor: Peter A. Troch, Ph.D.
- Jan. **Doctor of Philosophy**, *University of Sao Paulo - Sao Carlos School of Engi-
2020–Nov. neering*, Title: [Advances in open source hydroinformatics for flood modeling and
2023 disaster education](#) (GPA 4.0). Concentration area: Hydraulic Engineering and
Sanitation, [PhD thesis recently awarded as the best of 2023 in all engineering
fields of USP](#), also news published [at USP website](#).
Advisor: Eduardo M. Mendonzo, Ph.D. Co-advisor: Marcio H. Giacomoni, Ph.D., P.E.
- Jan. **Doctor of Philosophy**, *University of Texas at San Antonio*, Concentration
2020–present area: Civil and Environmental Engineering, Title: Real-time Control of Green-
Infrastructure and Stormwater Drainage Systems Through Looped-Feedback Con-
trol Approaches. Status: Graduation on hold (1 credit missing, currently on hold
due to my current postdoc position and to only be able to defend afterward due
to J1 visa constraints).
Advisor: Marcio Hofheinz Giacomoni, Ph.D., P.E.
- Feb. **Master's in Hydraulic Engineering and Sanitation**, *University of Sao Paulo
2018–Dec. - Sao Carlos School of Engineering.*, Concentration area: Hydraulic Engineering
2019 and Sanitation
Status: Graduated. Advisor: Eduardo M. Mendonzo.
- Feb. **B.Sc. in Civil Engineering**, *State University of Maringa - Brazil*
2013–Nov. Advisor: Germano Romera, Ph.D.
2017

Professional Interests

Physics-based hydrologic/hydrodynamic modeling; data assimilation & remote sensing; drought/flood
resilience; decision-support for water security under land-use & climate change, Physics Informed
Machine Learning, Optimization.

Supervision

Master's **Dr. Eduardo Mario Mendonzo**
Degree

Ph.D. Degree **Dr. Marcio Hofheinz Giacomoni**, *Dr. Eduardo Mario Mendondo*
Postdoctorate **Dr. Peter Troch**
Postdoctorate **Dr. Steve Gorelick**

Skills and Activities

Programming Language Matlab, Python, VBA, Javascript
Environment Latex, Microsoft Office, Git, Github, HTML, Wordpress, Inkscape, Google Earth Engine, QGIS
Operating System Windows, MacOS, Linux (basic)

Current Enrolled Courses

○ CS 229 - Machine Learning, Stanford University <https://cs229.stanford.edu/>.

Pedagogical Development

CIRTL Completed CIRTL program course through the University of Arizona, focused on evidence-based teaching and inclusive learning practices.

Work Experience

Feb. 2026–present **Post-doctoral Researcher**, *Stanford University - Doerr School of Sustainability, Department of Earth System Science*
Task: Using advanced remote sensing algorithms coupled with physically-based fully distributed hydrologic-hydrodynamic models, the main task of my work is to reconstruct all recent floods ($\approx$ last 20 years) in India and California. Following, machine learning inference models will be designed to identify the relationship of floods with millions of medical records.

Jan. 2024–Jan. 2026 **Post-doctoral Research Associate**, *University of Arizona - Department of Hydrology and Atmospheric Sciences*
Task: At the [Landscape Evolution Observatory](#), coupled a simplified 1D vadose-zone model with a 2D Boussinesq groundwater model and particle-filter data assimilation to identify state-dependent effective material properties. At Beaver Creek, developed a hillslope-storage Boussinesq catchment model with advanced snow representation to estimate hydrologic signatures.

2024-Spring **Teaching Assistant of Fundamentals of Surface Water Hydrology**, *University of Arizona - Department of Hydrology and Atmospheric Sciences*
Task: Course taught by Peter A. Troch, and I was responsible for reviewing and grading all assignments, projects, and exams.

Jan. 2023–Dec. 2023 **Graduate Research Assistant**, *School of Civil and Environmental Engineering, and Construction Management, Department of Civil and Environmental Engineering*
Task: Perform stormwater water quality analysis, develop numerical models, and set up and monitor field experiments. I was responsible for setting up autosamplers, rain gauges, v-notch weirs, and collecting and analyzing samples after storm events. In the water quality analysis, I was responsible for performing total suspended solids, volatile suspended solids, heavy metals, bacteria, pH, and temperature analyses. In this project, I was also responsible for the hydrological and hydraulic design of different permeable pavement structures funded by the City of San Antonio and constructed in January-2023 and under monitoring.

- 2022–2023 **Teaching Assistant of Hydrology I and II for Civil Engineering**, *University of Sao Paulo - Sao Carlos School of Engineering*
Task: Course taught by Eduardo M. Mendonado and I was responsible for developing supporting materials, spreadsheets and numerical models for the applied hydrology classes, as well as teaching some of the classes. Classes were offered for students in the Junior and Sophomore years.
- Feb. **Graduate Research Assistant**, *University of Sao Paulo - Sao Carlos School of Engineering, Department of Hydraulic Engineering and Sanitation*
 2018–Dec. **Task:** Perform stormwater water quality analysis, develop numerical models, and set up and monitor field experiments. I was also responsible for planning, designing, and controlling the execution of a bioretention system funded by FAPESP grants 2017/21940-0 and 2018/20865-0
- 2018–2019 **Teaching Assistant of Hydrology I and II for Civil Engineering**, *University of Sao Paulo - Sao Carlos School of Engineering*
Task: Develop supporting materials, spreadsheets, and numerical models for Applied Hydrology classes. Classes were offered for students in the Junior and Sophomore years.
- June **Undergraduate Research Assistant**, *State University of Maringa - Civil Engineering*,
 2016–June Brazil
 2017 **Task:** Conducted research on flood mitigation in downtown Umuarama and proposed nature-based solutions to reduce urban inundation impacts.
- 2015–2017 **Instructor of Financial Mathematics**, *CIEE (Center for School-Business Integration, Brazil)*
Task: Taught financial mathematics in the youth apprenticeship program.
- 2015–2017 **Instructor of Mathematics, Physics, and CAD-Assisted Technical Drawing**, *Yesbras*
Task: Taught mathematics, physics, and technical drawing with CAD support.
- 2014–2016 **Teaching Assistant**, *State University of Maringa*
Task: Help, teach, and solve tests, exams, and proposed exercises from Physics I and Physics II of Civil Engineering and Environmental Engineering Classes. Classes were offered for students in the Sophomore year.
- 2013–2017 **Teaching Assistant**, *Curso e Colégio Alfa Preparatory School*
Task: Help, teach, and solve pre-school exams of Physics and Mathematics for students in preparation for college.

Extra Activities

I run a [website](#) and a [Youtube](#) channel with content of Civil and Environmental Engineering (In Portuguese), and I plan to translate them to English in the near future. I developed over 60 numerical tools to aid in various engineering calculations, including the design of pipes, concrete structures, and the solution of hydrology and hydraulics problems. The website has garnered more than 90,000 downloads from over 10 countries, which has allowed me to network with designers and engineers worldwide. My goal is to provide high-quality, useful content to the engineering and water resources community, free of charge in most cases. I also write online articles (+30) about water resources and civil engineering overall. The articles are the main source of leads to the website and are top-ranked in Brazilian search engine optimization. The website is available at: www.engenheiroplanilheiro.com.br (In Portuguese).

Scholarships

- Ph.D Degree at UTSA Financed by the City of San Antonio - Texas (\$ 153,000 USD). Project - [Demonstrating the Environmental Benefits of Permeable Paved Surfaces over the Edwards Aquifer](#)
- Moore Foundation Offered Gordon and Betty Moore Foundation scholarship/fellowship support for 2025–2026 (approximately \$70,000); not accepted.
- Ph.D. Degree at USP Ph.D. scholarship by [CAPES](#) (Grant 88887.663281/2022-00; R\$ 74,400)
- M.Sc. Degree Master's degree scholarship financed by [FAPESP](#) (Process 18/20865-0; R\$ 58,116)

FAPESP Regular Research Grant	Project 2017/21940-0, 2018–2020, R\$ 158.243,56
Bsc. Degree	Free tuition by state university scholarship (approximately R\$ 105,000.00).

Fields and Keywords of Interest

Civil and Environmental Engineering	Flood Control, Water Quality Transport and Fate, Hydrodynamic Models, Hydrological Models, Water Quality Models, Low Impact Development, System Analysis, Urban Drainage, Disaster Risk Reduction, Water Resources Planning and Management, Closure-relationships, Soil Physics, Sub-surface Hydrology
Electrical Engineering	Linear Systems and Control, Control of Non-Linear Systems, Model Predictive Control, Linear Quadratic Regulators, Model Predictive Control.
Mathematics, Computer Science	Numerical Modeling, Explicit Methods, Implicit Methods, Finite-Difference Methods, GPU Processing, Convex Optimization, Non-linear Non-Convex Optimization, Data Assimilation, Reinforcement Learning, Particle Filter

Reviewer for Scientific Journals

2020-present	<i>Journal of Hydrologic Engineering</i>
2020-present	<i>Revista Brasileira de Recursos Hídricos</i>
2020-present	<i>Revista DAE</i>
2021-present	<i>Journal of Hydrology</i>
2021-present	<i>Journal of Water Resources Planning and Management</i>
2021-present	<i>Revista REGA</i>
2022-present	<i>Hydrological Sciences Journal</i>
2022-present	<i>Sustainable Cities and Society</i>
2023-present	<i>IEEE Transactions on Control of Network Systems</i>
2024-present	<i>Urban Water Journal</i>
2024-present	<i>The International Journal of River Basin Management</i>
2024-present	<i>Journal of Flood Risk Management</i>
2024-present	<i>Scientific Reports</i>
2024-present	<i>Geomatics - Natural Hazards and Risk</i>
2024-present	<i>AIMS Geosciences</i>
2024-present	<i>Journal of Asian Architecture and Building Engineering</i>
2025-present	<i>Hydrology and Earth System Sciences</i>
2025-present	<i>Geosciences</i>
2025-present	<i>Applied Sciences</i>
2025-present	<i>Water Resources Research</i>
2025-present	<i>Natural Hazards and Earth Systems Sciences</i>
2025-present	<i>Water Research - X</i>
2025-present	<i>Hydrology</i>
2025-present	<i>Applied Sciences</i>
2025-present	<i>Environmental Modelling and Software</i>
2025-present	<i>Case Studies in Construction Materials</i>
2025-present	<i>Revista Ambiental e Sanitária</i>
2026-present	<i>International Journal of Disaster Risk Reduction</i>
2026-present	<i>Environmental and sustainability indicators</i>

Peer Review Referee Activities (ORCID, Jan 2026)

Performed a total of **73** verified peer reviews for **27** manuscripts and grant submissions in international journals, including *Applied Sciences* (2), *Case Studies in Construction Materials* (4), *Engenharia Sanitária e Ambiental* (1), *Environmental and Sustainability Indicators* (2), *Environmental Modelling & Software* (3), *Geomatics, Natural Hazards & Risk* (1), *Geosciences* (1), *Hydrology* (3), *Hydrology and Earth System Sciences* (1), *IEEE Transactions on Control of Network Systems* (3), *International Journal of Disaster Risk Reduction* (2), *Journal of Flood Risk Management* (2), *Journal of Hydrologic Engineering* (10), *Journal of Hydrology* (11), *Natural Hazards and Earth System Sciences* (1), *RBRH – Revista Brasileira de Recursos Hídricos* (3), *Revista de Gestão de Água da América Latina* (5), *Scientific Reports* (2), *Sustainability* (1), *Urban Water Journal* (2), *Water* (5), *Water Research* (6), and *Water Resources Research* (2).

Research Metrics

ResearchGate

- **Research Interest Score:** 526.7
- **Citations:** 654
- **h-index:** 14
- **Profile:** [Research Gate](#)

Google
Scholar

- **Citations:** 709 (since 2021: 702)
- **h-index:** 14
- **i10-index:** 16
- **Updated:** July 19, 2026
- **Profile:** [Google Scholar](#)

Professional Memberships

- American Society of Civil Engineers (ASCE) - 2020-2021
- American Geophysical Union (AGU) - 2021-2022, 2024-present
- Brazilian Water Resources Association (ABRH) 2019-2020, 2021-2022.

Professional Service

2025 Panelist reviewer for *El Día del Agua y la Atmósfera*, University of Arizona.

Publications in peer-reviewed journals - Published or Accepted

Notation (*) indicates undergraduate student first author under my mentorship.

2026

1. Sánchez, M. H. (*), **Gomes Jr., M. N.**, Castillo, L. M. R., Castro, M. A. R. A., and Mendiondo, E. M. (2026). *Assessment of urban vulnerability to floods under current and future demographic and climate scenarios*. **International Journal of Disaster Risk Reduction**, 106337. DOI: [10.1016/j.ijdr.2026.106337](https://doi.org/10.1016/j.ijdr.2026.106337).
2. **Gomes Jr., M. N.** (2026). *LID-Design Atlas: Mapping Hydrologic Performance and Cost Trade-Offs in Infiltration-Based LID Design*. **Journal of Hydrologic Engineering**, 31(5). DOI: [10.1061/JHYEFF.HEENG-6901](https://doi.org/10.1061/JHYEFF.HEENG-6901).
3. Navarro, F. A. R., **Gomes Jr, M. N.**, & Mendiondo, E. M. (2025). *Risk management framework for urban catchments under non-stationary analysis: Mitigation and adaptation scenarios in a Brazilian case study*. **Urban Water Journal** (Accepted).

4. **Gomes Jr, M.N.**, 2026. Accounting for spatial runoff variability in LID design for urban catchments: model and software development. **Brazilian Journal of Water Resources**, 31, p.e10. DOI: <https://doi.org/10.1590/2318-0331.312620250060>
5. **Gomes Jr., M. N.**, Bauser, H. H., Troch, P. A. *Identifying the State Dependence of Effective Material Properties in a Simplified Hydrologic Hillslope Model*. **Water Resources Research**, 62(6), e2025WR041468, 2026. DOI: <https://doi.org/10.1029/2025WR041468>.
6. Brandão, A.R., Castro, M.A., Sánchez, M.H., **Gomes Jr, M.N.**, Uchôa, J.G.S., Vaz, I.C., Ghisi, E., Anache, J.A., Wendland, E.C., Oliveira, P.T. and Mendiondo, E.M., 2026. Flood Event Escalation and Urban Drainage Design Implications Under Nonstationary Rainfall in São Paulo State, Brazil. **Water**, 18(5), p.561. DOI: <https://doi.org/10.3390/w18050561>
7. Castro, M., **Gomes Jr., M. N.**, Montenegro, S. (2026) *Assessing the Socioeconomic Benefits of the Agreste Water Supply System in Pernambuco, Brazil*. **Brazilian Journal of Water Resources**. DOI: <https://doi.org/10.1590/2318-0331.312620250154>.
8. Brasil, J., Papagiannakis, A. T., Johnson, D. W., **Gomes Jr., M. N.**, and Giacomoni, M. H. (2026). *Field Scale Test Bed for Assessment of the Surface Runoff and Water Quality Performance of Permeable Pavements*. **Journal of Hydrologic Engineering**, 31(4), p.04026014. DOI: <https://doi.org/10.1061/JHYEFF.HEENG-6602>.
9. Sousa, M. R. (*), **Gomes Jr., M. N.**, Mendiondo, E. M. *Hydrological–Hydrodynamic Modeling of Climate-Induced Urban Flooding of Design Storms Using HydroPol2D: A Case Study in São Carlos, Brazil*. **Brazilian Journal of Water Resources**. DOI: <https://doi.org/10.1590/2318-0331.312620250090>.

2025

10. **Gomes Jr, M.N.**, Castro, M.A., Castillo, L.M., Sánchez, M.H., Giacomoni, M.H., de Paiva, R.C. and Bates, P.D., 2025. *Spatio-temporal performance of 2D local inertial hydrodynamic models for urban drainage and dam-break applications*. **Journal of Hydrology**, p.134661. DOI: <https://doi.org/10.1016/j.jhydrol.2025.134661>
11. Rápalo, L. M. C., **Gomes Jr., M. N.**, Mendiondo, E. M. *Multiple Degrees of Human Instability Due to Urban Overland Flow Within the 21st Century: An Urban Watershed Case Study in Brazil*. **International Journal of Disaster Risk Reduction**. DOI: <https://doi.org/10.1016/j.ijdrr.2025.105931>.
12. **Gomes Jr., M. N.**, Brasil, A., Johnson, D. W., Papagiannakis, A. T., Giacomoni, M. H. (2025) *A Coupled Darcy–Richards Framework for Hydrological Modeling of Permeable Pavements, Green Roofs, and Bioretention Systems*. **Environmental Modelling and Software**. DOI: <https://doi.org/10.1016/j.envsoft.2025.106766>.
13. **Gomes, M.N.**, Jalihal, V., Castro, M. and Mendiondo, E.M., 2025. *Exploring the impact of rainfall temporal distribution and critical durations on flood hazard modeling*. **Natural Hazards**, 121(9), pp.10989–11012. <https://doi.org/10.1007/s11069-025-07186-3>.
14. **Gomes Jr, M.N.**, do Lago, C.A. and Mendiondo, E.M., 2025. *A Simple Method for Designing Infiltration Low Impact Development Techniques Considering Effects of Urbanization and Climate Change*, **Brazilian Journal of Water Resources (RBHR)** (2025). DOI: <https://doi.org/10.1590/2318-0331.302520240106>
15. Vasconcelos, J.G., **Gomes Jr, M.N.**, Oliveira, P.T.S., Yang, D. and Fang, X., 2025. *Reformulating the Rational Method Considering Dissimilar Land Use Types*. **Journal of Irrigation and Drainage Engineering**, 151(4), p.04025017. <https://doi.org/10.1061/JIEDDH.IRENG-10431>.

2024

16. **Gomes Jr**, Marcus Nóbrega, Ahmad F. Taha, Luis Miguel Castillo Rápalo, Eduardo Mario Menciondo, and Marcio Hofheinz Giacomoni. “Real-time regulation of detention ponds via feedback control: Balancing flood mitigation and water quality.” **Journal of Hydrology** 643 (2024): 131866. DOI: <https://doi.org/10.1016/j.jhydrol.2024.131866>.
17. **Gomes Jr**, Marcus Nóbrega, Maria de Andrade Rocha Alencar Castro, Pedro Gustavo Câmara da Silva, Marcio Hofheinz Giacomoni, and Eduardo Mario Menciondo. *Increasing flood awareness through dam-break serious games*. **International Journal of Disaster Risk Reduction** 108 (2024): 104543. DOI: <https://doi.org/10.1016/j.ijdrr.2024.104543>.
18. Rápalo, Luis MC, Marcus N. **Gomes Jr**, and Eduardo M. Menciondo. “Developing an open-source flood forecasting system adapted to data-scarce regions: A digital twin coupled with hydrologic-hydrodynamic simulations.” **Journal of Hydrology** (2024): 131929. DOI: <https://doi.org/10.1016/j.jhydrol.2024.131929>.
19. Ambrogi Ferreira Do Lago, Cesar, Jose Artur Teixeira Brasil, Marcus Nóbrega **Gomes**, Eduardo Mario Menciondo, and Marcio H. Giacomoni. “Improving pluvial flood mapping resolution of large coarse models with deep learning.” **Hydrological Sciences Journal** 69, no. 5 (2024): 607–621. DOI: <https://doi.org/10.1080/02626667.2024.2329268>.

2023

20. M. N. **Gomes Jr.**, C. A. F. do Lago, L. M. C. Rápalo, P. T. S. Oliveira, M. H. Giacomoni, and E. M. Menciondo, “HydroPol2D – Distributed Hydrodynamic and Water Quality Model: Challenges and Opportunities in Poorly-Gauged Catchments,” **Journal of Hydrology**, 2023. DOI: <https://doi.org/10.1016/j.jhydrol.2023.129982>.
21. M. N. **Gomes Jr.**, L. M. C. Rapalo, P. T. S. Oliveira, M. H. Giacomoni, C. A. F. do Lago, and E. M. Menciondo, “Modeling unsteady and steady 1-d hydrodynamics under different hydraulic conceptualizations: Model/software development and case studies,” **Environmental Modeling and Software**, 2023. DOI: <https://doi.org/10.1016/j.envsoft.2023.105733>.
22. M. N. **Gomes Jr.**, M. H. Giacomoni, M. B. de Macedo, C. A. F. do Lago, J. A. Teixeira, T. R. Pereira, and E. M. Menciondo, “A modeling framework for bioretention analysis: Assessing the hydrologic performance under system’s uncertainty,” **Journal of Hydrologic Engineering**, DOI: <https://doi.org/10.1061/JHYEFF.HEENG-5705> 2023.
23. A. S. Ballarin, J. G. S. M. Uchôa, M. S. dos Santos, A. Almagro, I. P. Miranda, P. G. C. da Silva, M. N. **Gomes Jr.**, E. Wendland, P. T. S. Oliveira, “Brazilian water security threatened by climate change and human behavior”. **Water Resources Research**, DOI: <https://doi.org/10.1029/2023WR034914> 2023.
24. do Lago, C.A., Giacomoni, M.H., Bentivoglio, R., Taormina, R., **Junior**, M.N.G. and Menciondo, E.M., 2023. *Generalizing rapid flood predictions to unseen urban catchments with conditional generative adversarial networks*. **Journal of Hydrology**, 618, p.129276, DOI: <https://doi.org/10.1016/j.jhydrol.2023.129276>.

2022

25. M. N. **Gomes Jr.**, M. H. Giacomoni, A. F. Taha, and E. M. Menciondo, “Flood risk mitigation and valve control in stormwater systems: State-space modeling, control algorithms, and case studies,” **Journal of Water Resources Planning and Management**, vol. 148, no. 12, p. 04022067, 2022. [Online]. Available: [10.1061/\(ASCE\)WR.1943-5452.0001588](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001588)
26. M. Batalini de Macedo, M. Nobrega **Gomes Júnior**, T. R. Pereira de Oliveira, M. H. Giacomoni, M. Imani, K. Zhang, C. Ambrogi Ferreira do Lago, and E. M. Menciondo, “Low impact development practices in the context of united nations sustainable development goals: A new concept, lessons learned and challenges,” **Critical Reviews in Environmental Science and Technology**, vol. 52, no. 14, pp. 2538–2581, 2022, DOI: <https://doi.org/10.1080/10643389.2021.1886889>

27. J. A. T. Brasil, M. B. de Macedo, T. R. P. de Oliveira, F. G. Ghiglieno, V. C. B. de Souza, G. Marinho e Silva, M. N. **Gomes Júnior**, F. A. A. de Souza, and E. M. Menciondo, “Can we scale digital twins of nature-based solutions for stormwater and transboundary water security projects?” **Journal of Hydroinformatics**, 2022, DOI: <https://doi.org/10.2166/hydro.2022.142>
28. M. Batalini de Macedo, M. N. **Gomes Júnior**, V. Jochelavicius, T. R. P. de Oliveira, and E. M. Menciondo, “Modular design of bioretention systems for sustainable stormwater management under drivers of urbanization and climate change,” **Sustainability**, vol. 14, no. 11, p. 6799, 2022, <https://doi.org/10.3390/su14116799>.
29. Jochelavicius, V., E. M. Menciondo, M. N. **Gomes Jr.**, and M. B. de Macedo, “Construction of Intensity-Duration-Frequency curves with future climate change scenarios for the city of São Carlos - SP aiming at the design of compensatory techniques,” **Revista DAE**, vol. 147, p. 19-32, DOI: <https://doi.org/10.36659/dae.2022.065> (In Portuguese).

2021

30. J. Brasil, M. Macedo, C. Lago, T. Oliveira, M. **Júnior**, T. Oliveira, and E. Menciondo, “Nature-based solutions and real-time control: Challenges and opportunities,” **Water**, vol. 13, no. 5, p. 651, 2021, DOI: <https://doi.org/10.3390/w13050651>.
31. T. R. P. de Oliveira, M. B. de Macedo, T. H. Oliveira, C. A. F. do Lago, M. N. **Gomes Jr.**, J. A. T. Brasil, and E. M. Menciondo, “Different configurations of a bioretention system focused on stormwater harvesting in Brazil,” **Journal of Environmental Engineering**, vol. 147, no. 12, p. 04021058, 2021, [https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0001938](https://doi.org/10.1061/(ASCE)EE.1943-7870.0001938)

2020

32. M. N. **Gomes Jr.**, P. H. B. Alves, E. M. Menciondo, and L. F. R. Reis, “Statistical, visual and non-parametric analyses for the optimization of IDF curve fitting and construction of hydraulic works design abacuses: case study in São Carlos - SP”. **Revista DAE**, DOI: <https://doi.org/10.36659/dae.2021.013> 2020 (In Portuguese).
33. K. McClymont, D. G. F. Cunha, C. Maidment, B. Ashagre, A. F. Vasconcelos, M. B. de Macedo, M. F. N. Dos Santos, M. N. G. **Júnior**, E. M. Menciondo, A. P. Barbassa et al., “Towards urban resilience through sustainable drainage systems: A multi-objective optimisation problem”, **Journal of Environmental Management**, vol. 275, p. 111173, 2020, DOI: <https://doi.org/10.1016/j.jenvman.2020.111173>

Publications - Under Review or In Preparation

First-author manuscripts

1. **Gomes Jr., M. N.**, Castro, M. A. R. A., Nguyen, A., Wang, A., Gorelick, S. M. *Global Inequalities in Urban Hazard Risk*. **Nature Communications** (In preparation).
2. **Gomes Jr., M. N.**, M. A. R. A. Castro. *Continental-scale XGBoost Framework for Multi-Hazard Susceptibility and Socioeconomic Exposure in South America*. **Sustainable Cities and Society** (Under Review).
3. **Gomes Jr., M. N.**, Castro, M., Ballarin, A., Troch, P., Almagro, A., and Oliveira, P. T. S. *Levels and Indicators for Monitoring Inundation Across Rivers (LIMIAR): Hydraulic Flood Reconstruction in Brazil*. **Hydrology and Earth System Sciences** (Under preparation).
4. **Gomes Jr., M. N.**, Ballarin, A. S., Oliveira, P. T. S. *Time Distribution of Heavy Rainfall in Brazil: Empirical Huff Curves from 290,164 Sub-Daily Storm Events*. **Water Resources Research** (Under Review).

5. **Gomes Jr., M. N.** *A Two Gamma Reduced Analytical Hydrograph for SCS-CN Huff Design Storms.* **Journal of Hydrologic Engineering** (Under Review).
6. **Gomes Jr., M. N.** *A Nationally Consistent Assessment of Intensity–Duration–Frequency Curves for Brazil Using Long-Term Gridded and Satellite Rainfall Data.* **Physics and Chemistry of the Earth** (Under Review).

Coauthored manuscripts

7. Castro, M. A. R. A., **Gomes Jr., M. N.**, Mendiondo, E. M. *Probabilistic Dam Break Flood Mapping via Monte-Carlo Simulation.* **Journal of Hydrology.**
DOI: <https://essopenarchive.org/doi/full/10.22541/essoar.175181464.43589172> (Submitted).
8. Castro, M. A. R. A., **Gomes Jr., M. N.**, Mendiondo, E. M. *Trade-Offs Between Full Momentum, Diffusive Wave, and Local-Inertial Solvers for Dam-Break Flood Modeling.* **Environmental Modelling & Assessment** (Under review).
9. Brasil, J. T., **Gomes Jr., M. N.**, Johnson, D. W., Papagiannakis, A. T., Mendiondo, E. M., Giacomoni, M. H. *A Digital Twin Framework for Real-Time Probabilistic Forecasting of Permeable Pavement Performance.* **Journal of Hydrology** (Under Review).

Selected Open-Source Models Developed (7 of 24 public repositories on GitHub)

Name	Description	Public links
HydroPol2D	Distributed 2D hydrologic-hydrodynamic and water-quality model for overland flow, infiltration, groundwater recharge, shallow groundwater, reservoirs, dam-breaks, snow, and pollutant transport.	GitHub: repo Docs: site
MIRA	Interactive South America multi-hazard platform for flood, drought, wildfire, socioeconomic context, and city-priority analysis.	Viewer: site
GRIDF	National toolbox for gridded, bias-corrected IDF curves in Brazil, with diagnostics and a Google Earth Engine visualization app.	GitHub: repo App: gridf-br
Huff Curves BR Atlas	Interactive atlas of empirical Huff curves for ANA sub-daily rainfall stations in Brazil, including station search, quartile diagnostics, and design-storm export tools.	Viewer: site
LIMIAR	River-flood reconstruction system for ANA gauge stations and hydraulic flood reconstruction in Brazil.	GitHub: repo Docs: site
DRAIN-LID	Richards-based infiltration, evaporation, drainage, and pollutant/mass-balance model for permeable pavements, green roofs, and bioretention systems.	GitHub: repo Docs: site
India Flood Inventory Toolbox (IFI)	Toolbox for India Flood Inventory analysis, district-scale flood records, and related dashboard tools.	GitHub: repo Dashboard: site

Publications in Conference Proceedings

Notation (*) indicates undergraduate student first author under my mentorship.

Full articles in conference proceedings

27. Brasil, J.A., Papagiannakis, A.T., Giacomoni, M.H., Johnson, D., **Gomes Jr, M.N.** and Mendiondo, E.M., 2023. Hydrologic and Water Quality Treatment Performance of Four Permeable Pavement Surfaces. In *International Low Impact Development Conference 2023* (pp. 114-122). *Award of Best Paper*.
26. M. N. **Gomes Jr.**, Giacomoni, M.H., Taha, A.F. and Mendiondo, E.M., 2022, August. *Model Predictive Control for Stormwater Reservoirs: Investigating Effects of Climate Change and Urbanization*. **IEEE Conference on Control Technology and Applications (CCTA)** (pp. 691-698). IEEE.
25. M. N. **Gomes Jr.**, P. A. Braga, M. B. de Macedo, E. M. Mendiondo, "Preliminary design of detention ponds using specific design discharge and orifice stage discharge relationship for different climate patterns". **Brazilian Symposium of Water Resources, 2021**.
24. M. N. **Gomes Jr.**, P. A. Braga, E. M. Mendiondo, "Impact of rainfall changes on the Pacaembu-SP detention reservoir: assessment of water security regarding flood risk". **Brazilian Symposium of Water Resources, 2019** (In Portuguese).
23. M. N. **Gomes Jr.**, E. M. Mendiondo, F. Dornelles, A. T. Papagiannakis, "Permeable Pavement Hydrological Model to Assess the Long-Term Efficiency of Maintenance Using High-Resolution Temperature and Rainfall Data". **World Environmental and Water Resources Congress, 2021**, 1103-1117.
22. M. N. **Gomes Jr.**, E. O. Pavan, L. M. C. Rápallo, M. H. Giacomoni, E. M. Mendiondo, "Bidirectional Hydrodynamic and Water Quality Model (2DCAWQ): Modeling Challenges in Basins with Scarce Data: Application in the Tijuco Preto - São Carlos Basin". **XIV National Meeting on Urban Waters and IV Symposium on Revitalization of Urban Rivers, 2022** (In Portuguese).
21. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Mendiondo, "The role of raster resolution into overland flow and total suspended solids modeling in small urban catchments". **Brazilian Symposium of Water Resources, (2021)**.
20. M. N. **Gomes Jr.**, M. H. Giacomoni, A. T. Papagiannakis, "Spatial Assessment of Overland Flow,

Pollutant Concentration, and First Flush Using a 2D Non-Point Source Pollution and Hydrological Model for Urban Catchments". World Environmental and Water Resources Congress, 2021, 397-413.

19. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Menciondo, "*Bioretention Sizing via Process-Based Simulation: Generalized TC-Hydro Model Applied in Design Conditions*". **XIV National Meeting on Urban Waters and IV Symposium on Revitalization of Urban Rivers**, 2022 (In Portuguese).
18. M. N. **Gomes Jr.**, L. M. C. Rápalo, E. M. Menciondo, *A Serious Game for societal risk perception of dam-break flood assessment using a hydrodynamic model*, **Latin American Hydraulic Congress**, (2022). *Award of Best Paper*.
17. M. N. **Gomes Jr.**, C. A. F do Lago, M. H. Giacomoni, E. M. Menciondo, Assessing the role of cross-section shape for flood routing under compound flood events using a hydrodynamic model, **Latin American Hydraulic Congress**, (2022).
16. T. R. P. de Oliveira, M. N. **Gomes Jr.**, M. B. de Macedo, E. M. Menciondo, "*Análise da sensibilidade do potencial matricial em protótipo de biorretenção*". **Brazilian Symposium of Water Resources**, 2019 (In Portuguese).
15. CASTRO, M.A.R.A.; **Gomes Jr.**, M.N.; SANCHEZ, M.H.; FILHO, P.B.S., MENDIONDO, E.M. *Review of the Risk Index Applied to Dam Failure Simulation in Brazil*. **XXV Brazilian Symposium of Water Resources**. Anais. Aracaju (SE), 2023 (In Portuguese).
14. SANCHEZ, M.H; **Gomes Jr.**, M.N.; CASTRO, M.A.R.A.; MENDIONDO, E.M. *River flow forecasting methods: A review*. **XXV Brazilian Symposium of Water Resources**. Anais. Aracaju (SE), 2023.
13. M. R. de Sousa (*), M. N. **Gomes Jr.**, E. O. Pavan, L. M. C. Rápalo, F. A. R. Nararro, E. M. Menciondo, "*Physically Distributed Modeling Based on Urban Watersheds: Hydropol2D Model Applied to São Carlos (SP)*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
12. E. O. Pavan (*), M. N. **Gomes Jr.**, L. M. C. Rápalo, M. R. de Sousa, M. S. dos Santos F. A. R. Nararro, E. M. Menciondo, "*Physically Based Modeling of Large Basins - Challenges and Opportunities for Applying The Hydropol2D Model*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
11. L. F. Roysen (*), M. N. **Gomes Jr.**, E. O. Pavan, E. M. Menciondo, "*RTC-Stormwater Model: Advances In Flood Modeling And Control Of Urban Drainage Devices*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
10. A. S. Ballarin, J. G. S. M. Uchoa, M. S. dos Santos, A. Almagro, I. P. Miranda, P. G. C. da Silva, G. J. da Silva, M. N. **Gomes Jr.**, E. C. Wendland, P. T. S. de Oliveira, "*The Future of Water Security In Brazil: Separating The Impacts Of Climate Change And Water Demand*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
9. M. S. dos Santos, M. N. **Gomes Jr.**, D. A. Bressianim L. L. Ladeira, M. R. Benso, E. M. Menciondo "*Review of the Implementation of Model Predictive Control (MPC) In Urban Reservoirs*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
8. G. M. e Silva, M. R. Benso, P. G. C. da Silva, M. N. **Gomes Jr.**, E. M. Menciondo, "*Serious Game on Water Risks For Planetary Health: A New Model For Knowledge Co-Production*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023 (In Portuguese).
7. L. M. C. Rápalo, M. N. **Gomes Jr.**, D. A. Bressiani, M. R. Benso, E. M. Menciondo "*Threats of Urban Flash Floods on Human Stability for The Next Century*", **XXV Brazilian Symposium of Water Resources**. Proceedings. Aracaju (SE), 2023.
6. I. M. Benites, M. N. **Gomes Jr.**, A. Botari, J. C. Botari, L. Vanalli, "*Analysis of the Urban Drainage System: Case Study of the Stormwater Galleries at the intersection of Rua Governador Nei Braga and Avenida Brasil in the city of Umuarama-PR*". **XVII Safety, Health and Environment World Congress**, 2017 (In Portuguese).

Extended abstracts

5. M. N. **Gomes Jr.**, M. H. Giacomoni, E. M. Menciondo, A. F. Taha, "*Real-Time Control of*

Stormwater Reservoirs for Flood Risk Mitigation". The 2nd International Symposium on Water System Operations, 2021.

4. M. N. Gomes Jr., T. R. de Oliveira, T. H. Oliveira, M. B. de Macedo, M. H. Giacomoni, E. M. Mendiondo, "Nature-based solutions for sustainable stormwater: a model approach and sensitivity analysis for bioretention design using Green and Ampt and reservoir flood routing". **Second International Conference of Water, Megacities, and Global Change, UNESCO, 2021.**

Abstracts

4. Gomes Jr., M. N., Bauser, H. H., and Troch, P. A., *Characterizing the state dependence of effective material properties in a simplified hydrologic hillslope model*, **AGU24 Annual Meeting, Washington, DC, 2024.**
3. Gomes Jr. M. N., H. Bauser, P. A. Troch, *Identifying the state dependence of effective material properties in a simplified hydrologic hillslope model*, **The Conference on Computational Methods on Water Resources (CMWR), Tucson, 2024.**
2. M. N. Gomes Jr., C. A. F. do Lago, M. H. Giacomoni, E. M. Mendiondo, "Expanding the 2-Dimensional Green-Ampt and Non-linear Reservoir Hydrological Model from SWMM to MATLAB®". **AGU Fall Meeting, New Orleans, 2021.**
1. C. A. F. do Lago, M. N. Gomes Jr., E. M. Mendiondo, M. H. Giacomoni, "Application of Artificial Neural Networks to Predict Water Surface Elevation". **AGU Fall Meeting, 2021, New Orleans.**

Invited Talks, Short Courses, and Selected Presentations

1. **Research talk**, Stanford University, Stanford, California, Feb. 20, 2026. "Designing Open-Source Tools for Simulating Hydrological Processes: Challenges and Lessons from Building from Scratch."
2. **Invited research talk**, Landscape Evolution Observatory / Digital Twin Conference, University of Arizona, Tucson, Arizona, Dec. 11, 2025. "Identifying Effective Soil Properties in the Landscape Evolution Observatory."
3. **AGU24 Annual Meeting**, Washington, DC, Dec. 2024. Poster: "Characterizing the State Dependence of Effective Material Properties in a Simplified Hydrologic Hillslope Model."
4. **HydroPol2D short course**, University of Sao Paulo, Sao Carlos, Brazil, 2024. Model setup, running, and calibration.
5. **Invited talk, Hydraulic System Course**, University of Sao Paulo, Sao Carlos, Brazil, Apr. 1, 2023. Presentation on HydroHP-1D and HydroPol2D applications in Amazon basin and urban catchments.
6. **Invited undergraduate class presentation**, Vanderbilt University, Nashville, TN, Nov. 29, 2022. Two classes on RTC-Stormwater and real-time control of catchments, reservoirs, and channels.
7. **Latin American Hydraulic Congress**, Foz do Iguacu, Brazil, Nov. 2022. Presented two HydroHP-1D papers; one selected as best paper in its category.
8. **National Urban Water Meeting and Symposium on Revitalization of Urban Rivers**, Brasilia, Brazil, Sept. 2022. Presented two conference papers.
9. **Second International Conference Water, Megacities and Global Change**, UNESCO, Paris, France, Dec. 2021. Presentation on nature-based solutions and bioretention modeling.
10. **Brazilian Symposium of Water Resources**, Belo Horizonte, Brazil, Nov. 2021. Presentation on raster resolution and water-quality modeling in a fully distributed hydrologic-hydrodynamic model.
11. **2nd International Symposium on Water System Operations**, Bristol, UK, Sept. 2021. "Real-Time Control of Stormwater Reservoirs for Flood Risk Mitigation."
12. **World Environmental and Water Resources Congress**, Las Vegas, NV / online, June 2021. Presented papers on permeable pavement hydrology and distributed runoff/pollutant modeling.
13. **AGU Fall Meeting**, New Orleans, LA, Dec. 2021. "Expanding the 2-Dimensional Green-Ampt and Non-linear Reservoir Hydrological Model from SWMM to MATLAB."
14. **Advanced Excel short course**, University of Sao Paulo, Sao Carlos School of Engineering, Sao Carlos, Brazil, 2019. One-week course on advanced Excel methods for engineering calculations and data analysis.
15. **Brazilian Symposium of Water Resources**, Foz do Iguacu, Brazil, Dec. 2019. Presented papers on inverse modeling of the Pacaembu detention reservoir and orifice discharge sensitivity.

Grant Writing Collaboration

- **NSF Collaborative Research: Enabling Safe and Efficient Restoration of Drinking Water Systems through Multiphase Flow Modeling and Control** (2026–2029; Awards 2606145, 2606146, and 2606147; total collaborative award: \$749,667)
 - **PIs:** Marcio H. Giacomoni (University of Texas at San Antonio), Ahmad F. Taha (Vanderbilt University), and Jose G. Vasconcelos (Auburn University).
 - **Role:** Proposal contributor and project collaborator. Contributed the hydraulic conceptualization and governing equations for multiphase air–water filling and reviewed the modeling and control methodology.
- **HyREx: Enhancing Hydro-Resilience of Brazilian Catchments to Extreme Events in a Changing World** (FAPESP Process 2025/16429-3, 2026–2031; R\$ 520,368.25 + US\$ 20,700 + two undergraduate, two M.Sc., and one TT-IV fellowships)
 - **PI:** Andre Simoes Ballarin, University of Sao Paulo, Sao Carlos School of Engineering.
 - **Role:** Project collaborator on rainfall temporal structure, climate-change effects on flood extremes, and updated flood-design methods.
- **Funded project: Improving Continuous Streamflow Simulations for Beaver Creek, with Special Focus on Wet Winters** (Salt River Project, 2025; awarded \$129,342)
 - **PI:** Peter A. Troch; **Co-Investigator:** Patrick D. Broxton.
 - **Role:** Project researcher. Developed and calibrated the hsB-SM catchment model coupled with SnowPALM, prepared hydrometeorological and landscape data, and evaluated streamflow simulations with emphasis on wet winters.
- **Proposal submission: Improving Continuous Streamflow Simulations for Beaver Creek, with Special Focus on Wet Winters** (submitted to Salt River Project, 2026; proposed budget: \$80,000; not funded)
 - **PI:** Peter A. Troch; **Co-Investigator:** Patrick D. Broxton.
 - **Role:** Full proposal co-author with Peter A. Troch and Patrick D. Broxton. Co-wrote the technical narrative, preliminary analysis, and work plan for coupling SnowPALM with hsB-SM to improve streamflow simulation during wet winters.
- **Proposal development: Multi-Contaminant Exposure in Moab’s Backwaters under Climate Change with Integrated Surface–Groundwater and Transport Modeling** (developed for Keck Foundation consideration, 2025; estimated budget: \$1,010,660)
 - **PI:** Peter A. Troch.
 - **Role:** Proposal co-author and hydrologic–hydrodynamic modeler. Co-developed the monitoring and modeling plan coupling surface flow, river–aquifer exchange, and contaminant transport to estimate exposure to ammonia, uranium, salinity/TDS, and nitrate under historical and climate-change scenarios.
- **Analysis of Unit Hydrograph Models for San Antonio Watersheds** (approximate budget: \$49,000; not funded)
 - **P.I** - Dr. Marcio H. Giacomoni
 - **Tasks:** I helped to define the methods used to systematically assess the best unit hydrograph theory used for San Antonio observed data. The idea is that the standard PRF 484 unit hydrograph was oversizing the detention ponds in the city. Therefore, investigating the best-fitted unit hydrograph would reduce the associated costs of low-impact development designs.
- **Curbing Climate Change-Induced Floods via Control Theory for Urban Drainage Systems** (estimated budget: \$250,000; not funded)
 - **P.I’s** - Dr. Marcio H. Giacomoni and Dr. Ahmad F. Taha

- **Tasks:** Our previous [research](#) was the starting point of this proposal. We developed a real-time control model that can potentially increase the efficiency of stormwater systems for flood control. In this proposal, we plan to advance the mathematical modeling to include new stormwater systems (e.g., tunnels, dividers) and actuators (e.g., pumps and gates) to optimally control floods via real-time optimization.

Student Mentoring & Researcher Collaborations (selected)

- **Shikshita Bhandari** (Ph.D. student, Stanford University, 2026–pres.): flood effects on socioeconomics in India.
- **Alicia Nguyen** (Ph.D. student, Stanford University, 2026–pres.): composite hazards in India.
- **César A. F. do Lago** (Ph.D., UTSA/USP, 2018–pres.): hydrodynamic modeling, deep learning flood downscaling, ANN WSE prediction; co-developer in HydroPol2D applications.
- **Marina B. de Macedo** (Ph.D., UNIFEI/USP, 2018–pres.): LIDs, climate/urbanization impacts, modular bioretention design; multiple joint papers.
- **Luis M. C. Rápalo** (Ph.D., USP, 2021–2024): HydroPol2D chapters, urban flood forecasting digital twin, serious-games for risk perception.
- **Luis M. C. Rápalo** (Ph.D., USP, 2021–2025): flood inundation mapping with HydroPol2D. Flood damage economic analysis.
- **Ivan Cuervo** (M.Sc., UTSA, 2020–2022): field monitoring (autosamplers, weirs), water-quality analyses, 2-D runoff modeling.
- **Jalihal Vijay** (B.Sc., Vanderbilt, 2023–2024): flood modeling in Bellandur - India.
- **Artur A. T. Brasil** (Ph.D., UTSA, 2022–pres.): permeable pavements field study, digital twin forecasting for LIDs.
- **Mateo Hernandez** (M.Sc., USP, 2023–pres.): closed-loop flood control with distributed hydro–hydraulic model; HEC-RAS vs HydroHP comparisons.
- **Matheus dos Santos** (M.Sc., USP, 2022–2024): MPC in urban reservoirs using RTC-Stormwater; prediction-horizon sensitivity.
- **Maria A. R. A. Castro** (M.Sc./Ph.D., USP, 2023–pres.): Monte Carlo dam-break ensembles; trade-offs across LI/diffusive/full momentum solvers.
- **Enrico Pavan** (B.Sc., USP, 2022–2023): large-basin HydroPol2D setup; undergraduate research supervision.
- **Lucas F. Roysen** (B.Sc., USP, 2022–2023): real-time reservoir control; senior project supervision; RTC-Stormwater applications.
- **Milena Rosa** (B.Sc., USP, 2022–2024): HydroPol2D in data-scarce urban basin; pollutant fate/transport mapping, senior project supervision.
- **Vivian Jochelavicius** (B.Sc., USP, 2019): IDF curves under climate scenarios; design implications for compensatory techniques.

Current Collaborators

- | | |
|------------------------------------|---|
| Steven Gorelick,
Ph.D | Professor at Stanford University. Google Scholar , Website . |
| Peter Troch,
Ph.D | Professor at University of Arizona. Google Scholar , Website . |
| Hannes Bauser, Ph.D | Assistant Professor at University of Nevada, Las Vegas. Google Scholar , Website . |
| Paul D. Bates,
Ph.D | Professor at University of Bristol. Orcid , Google Scholar , Website . |
| Ahmad F. Taha, Ph.D | Associate Professor at Vanderbilt University. Google Scholar , Twitter , Website . |
| Eduardo Mario Mendiondo,
Ph.D | Professor at University of Sao Paulo, Sao Carlos School of Engineering. Orcid , Google Scholar . |
| José Vasconcelos,
Ph.D | Professor at Auburn University. Google Scholar . |
| Marcio H. Giacomoni,
Ph.D, P.E. | Professor at University of Texas at San Antonio. Orcid , Google Scholar , Website . |

- Paulo T.S. Oliveira, Ph.D. Associate Professor at Federal University of Campo Grande. [Orcid](#), [Google Scholar](#), [Twitter](#), [Website](#).
- Marina B. de Macedo, Ph.D. Associate Professor at Federal University of Itajubá. [Orcid](#), [Google Scholar](#), [Website](#).
- A. T. Papa-giannakis, Ph.D, P.E. Professor at University of Texas at San Antonio. [Orcid](#), [Google Scholar](#), [Website](#).
- César Ambrogi Ferreira do Lago, Ph.D. University of Texas at San Antonio. [Orcid](#), [Google Scholar](#).
- Daniele Bresisani, Ph.D. Associate Professor at Federal University of Pelotas. [Orcid](#), [Google Scholar](#), [Website](#).
- Luis Miguel Castillo Rápalo, Ph.D. University of Sao Paulo, Sao Carlos School of Engineering. [Orcid](#), [Google Scholar](#).
- Luis Miguel Castillo Rápalo, Ph.D. University of Sao Paulo, Sao Carlos School of Engineering. [Orcid](#), [Google Scholar](#).
- Fabricio A. Richmond Navarro , Ph.D. University of Sao Paulo, Sao Carlos School of Engineering. [Orcid](#), [Google Scholar](#).
- Tiantian Zhou, Ph.D. University of California, Davis. [Google Scholar](#).
- Andre S. Ballarin, Ph.D. Assistant Professor at University of Sao Paulo. [Orcid](#), [Google Scholar](#)

Press and Public Visibility

- Hydrologic models [Modelos hidrológicos desenvolvidos por alunos da USP ajudam a prever riscos de desastres](#) [English: *Hydrologic models developed by USP students help predict disaster risks*], **Jornal da USP**, Aug. 28, 2024.
- Public media [Alunos da USP criam ferramenta para prever desastres naturais](#) [English: *USP students create a tool to predict natural disasters*], **Olhar Digital**, Aug. 30, 2024; [Alunos da USP criam ferramenta que prevê desastres](#) [English: *USP students create a tool that predicts disasters*], **TudoCelular**, 2024.
- EESC-USP coverage [Novos modelos comunitários desenvolvidos por alunos ajudam com maior previsibilidade de riscos de desastres](#) [English: *New community models developed by students improve disaster-risk predictability*], EESC-USP, Aug. 6, 2024.
- Thesis award Coverage of the 13th *Prêmio Tese Destaque USP* [USP Distinguished Thesis Award] by **Jornal da USP**, **EESC-USP**, **EESC-USP result announcement**, **EESC-USP PPG**, and **PRPG-USP**, 2024.

Professional References

○ Steven Gorelick

- Professor at Stanford University
- gorelick@stanford.edu
- Department of Earth System Science
- [Google Scholar](#); [Website](#)

- **Peter A. Troch**
 - Professor at University of Arizona
 - patroch@arizona.edu
 - Hydrology and Atmospheric Sciences
 - [Google Scholar](#); [Website](#)
- **Paul D. Bates**
 - Professor at University of Bristol
 - paul.bates@bristol.ac.uk
 - School of Geographical Sciences
 - [Google Scholar](#); [Website](#)
- **Paulo T. S. Oliveira**
 - Associate Professor at the University of North Carolina at Charlotte
 - polivei2@charlotte.edu
 - Civil and Environmental Engineering
 - [Google Scholar](#); [Website](#)
- **Ahmad F. Taha**
 - Associate Professor at Vanderbilt University
 - ahmad.taha@vanderbilt.edu
 - Civil and Environmental Engineering
- **Marcio H. Giacomoni**
 - Professor at University of Texas at San Antonio
 - marcio.giacomoni@utsa.edu
 - College of Civil Engineering and Integrated Design
- **Eduardo M. Mendiondo**
 - Associate Professor at University of Sao Paulo, Sao Carlos School of Engineering
 - emm@usp.br
 - Hydraulic and Sanitation Engineering
- **Rodrigo Cauduro Paiva**
 - Associate Professor at the Federal University of Rio Grande do Sul
 - rodrigocdpaiva@gmail.com
 - Institute of Hydraulic Research